CBC HomeLearning Platform

AI-Powered Competency-Based Curriculum LMS

PRODUCT REQUIREMENTS DOCUMENT

Version 2.1 Merged | March 2026 | Draft for Development

Version	2.1 Merged (supersedes v2.0 Enhanced)
Date	March 2026
Status	Draft for Development
Scope	PP1 through Grade 12 — All Subjects — Kenya CBC/CBE
Prepared By	CBC HomeLearning Platform Development Team

Executive Summary

CBC HomeLearning Platform (v2.1 Merged) is an AI-powered Learning Management System for homeschooling families in Kenya, strictly aligned to the Kenya Institute of Curriculum Development (KICD) Competency-Based Curriculum (CBC) and Competency-Based Education (CBE) framework. This PRD merges v2.0 Enhanced with supplementary CBC/CBE context research to produce a single, comprehensive development reference.

This merged PRD addresses five critical capability gaps from v1.0 and additionally incorporates validated research on RAG-driven educational platforms, child-voice AI safeguards, and Kenyan-specific deployment constraints identified in the supplementary context document.

#	Capability	Description
1	RAG Architecture	Fully specified end-to-end RAG pipeline grounded in KICD curriculum documents; validated against published educational RAG research
2	Curriculum Content Generation	Lesson factory producing daily→weekly→termly content per KICD designs; 95%+ alignment target per research benchmarks
3	Voice AI & Conversational Tutoring	Bidirectional voice interaction using STT/TTS; child voice models trained on Kenyan accents; COPPA/FERPA/PDPA compliant
4	Timetable & Scheduling Engine	Automated multi-subject weekly timetables; AI-driven constraint-aware scheduling; iCal/PDF/WhatsApp export
5	CBC/CBE Assessment & Progress	8-level KNEC grading, competency portfolios, longitudinal tracking PP1–G12; no aggregate scores per KNEC directive

IMPORTANT — Merged Content Note: Sections marked with [+CBE Context] contain additions or enhancements derived from the supplementary Kenyan CBC/CBE Context document. All other content originates from PRD v2.0 Enhanced.

1. Kenya CBC/CBE Curriculum Structure

1.1 Education Philosophy [+CBE Context]

Kenya's revised basic education system (CBC/CBE) emphasises holistic competency development and values-based learning over rote memorisation. The framework is built on seven core competencies that must be developed across all grades and subjects:

- Communication and Collaboration
- Self-Efficacy
- Critical Thinking and Problem Solving
- Creativity and Imagination
- Citizenship
- Digital Literacy
- Learning to Learn

Assessment is continuous and descriptive: learners earn competency descriptors ("Exceeds Expectations", "Meets Expectations", etc.) for each outcome rather than aggregate percentage scores. All generated lessons, activities, and assessments must strictly align with official KICD curriculum for the correct grade and term, incorporating national values (responsibility, integrity, respect, love, unity, tolerance) and pertinent contemporary issues (PCIs) as mandated.

1.2 Education Levels & Subject Allocation (Post-2024 Rationalisation)

Following the Presidential Working Party on Education Reform (PWPER) recommendations, KICD rationalised subject counts at every level. The platform must support all tiers:

Level	Grades	Max Subjects	Key Learning Areas
Early Years Education	PP1–PP2	5	Language Activities, Mathematical Activities, Environmental Activities, Psychomotor & Creative Activities, Religious Activities
Lower Primary	Grades 1–3	7	Literacy (English), Literacy (Kiswahili), Mathematical Activities, Environmental Activities, Creative Arts, Physical & Health Ed, Religious Ed (CRE/IRE/HRE)
Upper Primary	Grades 4–6	8	English, Kiswahili, Mathematics, Integrated Science, Social Studies, Creative Arts & Sports, Religious Ed, Agriculture & Nutrition
Junior Secondary	Grades 7–9	9	English, Kiswahili, Mathematics, Integrated Science, Pre-Technical Studies, Social Studies, Business Studies, Creative Arts, Physical & Health Ed
Senior Secondary	Grades 10–12	7 (core + electives)	4 Compulsory Core + 3 Track Electives (STEM / Social Sciences / Arts & Sports / Vocational-Technical)

1.3 Assessment Framework — KNEC 8-Level CBC/CBE Grading

The platform's assessment engine must implement the official KNEC Competency-Based Assessment Framework (CBAF) 8-level scale. This applies to continuous School-Based Assessment (SBA) and all summative assessments.

Code	Descriptor	Score Range	Points	Meaning / AI Action
EE1	Exceeding Expectations 1	90–100%	8	Exceptional — applies knowledge creatively in novel contexts
EE2	Exceeding Expectations 2	75–89%	7	Very Good — consistently exceeds grade-level requirements
ME1	Meeting Expectations 1	58–74%	6	Good — fully meets expected competency standards
ME2	Meeting Expectations 2	41–57%	5	Fair — meets core competencies with minor gaps
AE1	Approaching Expectations 1	31–40%	4	Needs Improvement — progressing toward standard
AE2	Approaching Expectations 2	21–30%	3	Below Average — foundational gaps require targeted support
BE1	Below Expectations 1	11–20%	2	Significant gaps — intensive intervention recommended
BE2	Below Expectations 2	0–10%	1	Minimal engagement — full remediation pathway required

CRITICAL: No aggregate total scores. Each subject and competency is reported independently, per KNEC CBAF guidelines. The platform MUST NOT display or calculate mean scores across subjects. Reporting uses descriptive labels, not percentages, in all parent-facing dashboards.

2. RAG Architecture — Detailed Specification

2.1 RAG Design Principles [+CBE Context]

The RAG system is the backbone of curriculum fidelity. Every piece of content generated by any AI agent MUST be grounded in retrieved KICD curriculum document chunks. Research on RAG-powered educational platforms confirms this approach prevents hallucination by anchoring content in authoritative sources and enables rapid creation of lesson plans, quizzes, and assignments that match official standards.

Core RAG Guarantees:

- All generated lessons explicitly cite at least one KICD curriculum chunk by document ID, grade, subject, and strand
- Curriculum Alignment Agent validates every output against the retrieved chunks before delivery
- Hallucination guard: if confidence score from retrieval < 0.75, fallback to secondary retrieval before generation
- Monthly automated drift audits compare generated content embedding to source curriculum embeddings
- Version control on all curriculum documents; any KICD update triggers re-ingestion and cache invalidation
- 95%+ alignment target with official KICD objectives, validated per published RAG educational benchmarks [+CBE Context]

2.2 Knowledge Base Structure (Hierarchical Chunking Strategy)

The vector database must be populated with chunked KICD curriculum documents following a strict 5-level hierarchy. Curriculum PDFs are split into semantic chunks, embedded (multilingual-e5-large), and stored in Weaviate for on-demand semantic retrieval.

Chunk Level	Content	Key Metadata	Count Example
Level 0 — Document	Full curriculum design per subject+grade	grade, subject, term, KICD version	1 per subject/grade
Level 1 — Theme	Each thematic area (e.g. '1.0 The Family')	theme_id, theme_name, term, lesson_count	13 per Gr4 English
Level 2 — Strand	Listening/Speaking, Reading, Language Use, Writing	strand, sub_strand, lesson_count	4 per theme
Level 3 — Lesson Unit	Specific Learning Outcomes + Suggested Learning Experiences + Key Inquiry Questions	outcome_codes, PCIs, values, core_competencies	3–4 per strand
Level 4 — Assessment	Assessment criteria, rubric descriptors per SLO	assessment_type, competency_level, CBC_rubric	Per lesson unit

2.3 Adaptive Retrieval Strategy (Per Query Type)

Query Type	k Value	Retrieval Strategy	Chunk Types	Example
Simple Definition Query	k=2–3	Cosine similarity > 0.85	Single strand chunk	What is a determiner?
Lesson Plan Generation	k=5–8	Cosine > 0.78 + BM25 hybrid	Theme + strand + assessment chunks	Generate Lesson 3 on The Family

Query Type	k Value	Retrieval Strategy	Chunk Types	Example
Multi-Strand Daily Lesson	k=6–10	MMR diversity + similarity	All 4 strands for a theme	Full 40-min English lesson on Etiquette
Weekly Timetable Block	k=10–15	Cross-subject graph traversal	Multiple subjects + cross-subject links	Generate Week 3 timetable
Assessment Generation	k=4–6	Assessment-tagged chunks only	SLOs + rubric chunks	Create Term 1 English quiz
Weakness Remediation	k=3–5	Learner mastery profile filter	Previously failed outcome chunks	Student weak on adjectives — remediation plan

2.4 Knowledge Graph (Neo4j) — Curriculum Relationships

The knowledge graph encodes curriculum dependencies used for learning path sequencing and cross-subject integration. Key relationship types:

- PREREQUISITE_OF: Topic A must be mastered before Topic B (e.g., Nouns before Pronouns)
- STRAND_IN: Lesson Unit belongs to Strand in Theme in Grade in Subject
- ASSESSED_BY: Lesson Unit is assessed by specific CBC rubric criterion
- INTEGRATES_WITH: Cross-subject connection (e.g., English Letter Writing integrates with Social Studies Civics)
- DEVELOPS_COMPETENCY: Lesson Unit develops specific CBC core competency
- EMBEDS_VALUE: Lesson embeds CBC value (Responsibility, Respect, Unity, Love, Integrity, Tolerance)
- ADDRESSES_PCI: Lesson addresses Pertinent & Contemporary Issue (Digital Safety, Social Cohesion, Gender Education)
- SPIRALS_FROM: Upper-grade content that spirals from lower-grade foundation

2.5 RAG Pipeline — End-to-End Flow

Step	Trigger	Process	Owner Agent
1. Ingestion	KICD PDF/document upload	PDF parser → text extraction → hierarchical chunking → embedding (multilingual-e5-large) → Weaviate upsert + Neo4j node creation	Curriculum Admin
2. Query Formulation	Learner or system triggers content need	Context builder: learner_id + grade + subject + current_theme + mastery_profile → structured query	AI Orchestrator
3. Hybrid Retrieval	Query against vector DB	BM25 sparse retrieval + dense vector search → RRF fusion → re-rank by recency + learner mastery gap	RAG Engine
4. Graph Augmentation	Enrich with curriculum relationships	Neo4j: fetch prerequisite chain, cross-subject links, competency mapping → inject into context	Knowledge Graph
5. Context Assembly	Build LLM prompt context	Retrieved chunks + graph context + learner profile + CBC values/PCIs → structured system prompt	Context Builder
6. Generation	LLM produces curriculum content	Subject Agent generates lesson/assessment/timetable entry with mandatory citation of retrieved chunk IDs	Subject Agent
7. Validation	Curriculum Alignment Agent	Semantic similarity of output to source chunks (>0.75 threshold); CBC SLO coverage check; hallucination detection	Alignment Agent

Step	Trigger	Process	Owner Agent
8. Delivery & Cache	Serve content to learner	Cache validated content; log chunk citations; update learner mastery profile; trigger parent dashboard update	Delivery Layer

3. Curriculum Content Generation Engine

3.1 Lesson Factory — Daily→Weekly→Termly Pipeline

The Lesson Factory generates a complete academic year of structured content from raw KICD curriculum inputs. Using Grade 4 English as the MVP reference model, the factory operates at four nested levels:

Unit	Lessons	Duration	Coverage	Structure
Daily Lesson	1 lesson	40 minutes	1 theme + 1 strand + 1 SLO	Intro → Concept → Activity → Assessment check → Reflection
Weekly Block	5 lessons/week	200 minutes	1 theme, 4 strands covered	Mon–Fri with strand rotation (L&S → Read → Lang Use → Write → Review)
Theme Block	~12–14 lessons	~1 week/theme	All 4 strands for 1 theme	Full theme coverage with mid-theme and end-theme assessment
Termly Plan	~50 lessons/term	~10 weeks	4–5 themes per term	Term assessment + portfolio compilation + parent report
Annual Plan	~150 lessons/year	~30 weeks	All 13 themes (Gr4 English)	3 terms × 50 lessons; KPSEA/KJSEA prep in final term

3.2 Daily Lesson Structure Template (40 Minutes)

Every AI-generated lesson must conform to this CBC-aligned structure, populated strictly from retrieved KICD curriculum chunks:

- Header: Grade | Subject | Theme | Strand | Sub-strand | CBC Learning Outcome Code(s)
- Core Competency Focus: Which of the 7 CBC competencies are developed (must be explicit)
- Values Integration: CBC values embedded in the lesson (Responsibility, Respect, etc.)
- PCIs (Pertinent & Contemporary Issues): Any relevant PCIs addressed
- Key Inquiry Question: The KICD-specified inquiry question to anchor learning
- WARM-UP (5 min): Vocabulary activation / tongue twister / minimal pair activity
- CONCEPT INTRODUCTION (10 min): New knowledge delivery grounded in retrieved curriculum chunk
- GUIDED PRACTICE (12 min): Collaborative activity from Suggested Learning Experiences
- INDEPENDENT ACTIVITY (8 min): Learner performs task; AI monitors and provides scaffolded hints
- ASSESSMENT CHECK (3 min): 2–3 targeted oral/written questions mapped to SLO
- REFLECTION & CLOSURE (2 min): Learner self-assesses; AI provides formative feedback using CBC rubric language
- Resources: Suggested digital/physical materials (dictionary, audio, peer activity)
- RAG Citations: Chunk IDs from vector DB used to generate this lesson [MANDATORY]

3.3 Multi-Subject Content Generation (All CBC Grades) [+CBE Context]

The content generation system scales across all grades and subjects. The AI can automatically produce coherent lesson narratives, examples, practice questions, and summaries that precisely follow the CBC syllabus. Research confirms RAG-powered systems enable rapid creation of materials aligned to the curriculum at scale.

Subject Agent	Grades	Core Generation Focus	Notes
English Language Agent	PP1–G 12	Reading fluency, grammar, composition, functional writing	Grade 4 English CD is MVP reference
Kiswahili Agent	PP1–G 12	Language immersion, fasihi, sarufi, utungo, mazungumzo	Bilingual generation; Kiswahili-first for lower primary
Mathematics Agent	PP1–G 12	Step-by-step problem solving, visual models, number sense	Adaptive difficulty; spiral review from prior grade
Integrated Science Agent	G4–G9	Inquiry-based learning, experiment guidance, safety	Practical session generator for home experiments
Social Studies Agent	G4–G9	Geography, history, civics, current affairs	Map-based and timeline activities
Pre-Technical Studies Agent	G7–G9	Design process, basic engineering, ICT projects	Project-based practical sessions
Business Studies Agent	G7–G9	Entrepreneurship, basic bookkeeping, market survey	Simulation activities with Kenyan context
Agriculture & Nutrition Agent	G4–G6	Crop farming, soil, balanced diet, food safety	Seasonal Kenyan crop calendar integrated
Creative Arts & Sports Agent	PP1–G 12	Visual art, music, PE, drama project guidance	Portfolio-based; practical every session
Religious Ed Agent (CRE/IRE/HRE)	PP1–G 9	Values-based learning, ethical reasoning, scripture	Faith-track selection at onboarding
Senior Track Agents (STEM/SS/Arts/VT)	G10–G 12	Track-specific elective content	Activated at Grade 10 track selection

4. Voice AI & Conversational Tutoring

4.1 Voice Architecture Overview [+CBE Context]

Voice functionality is a first-class feature of the platform, not an add-on. For early learners (PP1–Grade 3) who may not yet read fluently, and for English/Kiswahili pronunciation activities at all grades, voice is the primary interaction mode. Research demonstrates that voice AI boosts literacy outcomes (e.g. listening to a child read and providing real-time feedback). The system implements a full bidirectional voice pipeline.

Voice Input (STT):

- Primary STT: Whisper (OpenAI) — multilingual, runs locally via whisper.cpp or API
- Fallback STT: Vosk lightweight offline model (50MB) for low-bandwidth; Web Speech API for browser
- Child voice model: trained on children's speech for accuracy with young learners [+CBE Context: trained on diverse Kenyan accents for fairness]
- Supports English and Kiswahili with Kenyan accent optimisation
- Voice activation: push-to-talk button (default) or wake-word 'Hey Tutor' (optional)
- Noise suppression for home/outdoor environments

Voice Output (TTS):

- Primary TTS: Kokoro-82M (open-source, runs locally on CPU); ElevenLabs API for premium voices
- Fallback TTS: Coqui TTS (open-source, offline, Kiswahili voice available)
- Child-friendly voice persona: warm, encouraging, age-appropriate pace
- Adjustable speed: slower for PP1–G2, normal for G3+, faster review mode
- Pronunciation model: curriculum vocabulary pre-registered with phonetic spellings
- Emotional tone: encouraging on correct answers, gentle and supportive on errors

LOW-BANDWIDTH NOTE [+CBE Context]: Given connectivity constraints in many Kenyan homes and schools, voice models must be cached locally. The app should use streaming cautiously and prioritise offline reliability. Offline STT (Vosk) and TTS (Coqui) must function without internet connectivity.

4.2 Conversational AI Tutor — Dialogue Patterns

The AI tutor conducts Socratic dialogue rather than direct answer-giving, in alignment with CBC's 'learning by doing' philosophy. The following dialogue modes are supported:

Dialogue Mode	When Used	Interaction Pattern	Technical Basis
Pronunciation Drill	L&S strand lessons	Tutor speaks word/phrase → Learner repeats → STT scores pronunciation → Tutor gives phonemic feedback	Whisper STT + CBC pronunciation rubric
Comprehension Q&A	Reading strand lessons	Tutor asks Key Inquiry Question → Learner answers (voice or text) → Tutor probes deeper using Socratic questioning	RAG-retrieved passage + KIQ from KICD CD
Grammar Practice	Language Use strand	Fill-in-the-blank via voice; tutor confirms with CBC rule explanation; learner constructs own sentence	Determiner/tense/word-class chunk from RAG
Composition Feedback	Writing strand	Learner dictates or types composition → Tutor gives	KICD writing rubric + CBC SLO

Dialogue Mode	When Used	Interaction Pattern	Technical Basis
		structured feedback: content, organisation, language use, grammar	
Quiz Mode	All strands	Voice-input MCQ and short answer; real-time scoring against CBC rubric; weakness flagging	Assessment Agent + Knowledge Tracing
Remediation Dialogue	Weak competencies	Tutor identifies knowledge gap → uses Socratic questioning to re-teach from different angle → re-assesses	Mastery-gap retrieval from knowledge graph
Encouragement Loop	All sessions	Adaptive praise calibrated to CBC level ('You are meeting expectations on...'); progress celebrations	Personalization Agent + streak tracker

VOICE SAFETY & PRIVACY: All STT transcripts of child speech are processed locally where possible. No child voice recordings are stored without explicit parental consent. Transcripts are anonymised for model improvement. Full compliance with Kenya Data Protection Act (2019), COPPA, FERPA, and GDPR principles required [+CBE Context].

5. Timetable & Scheduling Engine

5.1 Automated Timetable Generation [+CBE Context]

The Timetable Engine produces a structured, CBC-compliant weekly schedule from the learner's enrolled grade, selected subjects, and term plan. AI-driven timetabling considers constraints such as subject hours, breaks, and practical session requirements — research confirms this reduces manual scheduling effort from weeks to hours.

Timetable Generation Rules:

- Total lessons per week: 5 lessons/subject — per KICD allocation
- Session duration: 40 minutes per lesson (CBC standard)
- Core subjects (English, Kiswahili, Maths) scheduled in morning slots for peak cognitive performance
- Creative Arts, Physical Education, and practical sessions scheduled in the afternoon
- No more than 3 core academic subjects in sequence without a break or creative activity
- Practical sessions flagged and require minimum 60 minutes (double lesson slots)
- Weekly schedule exports as: PDF timetable, iCal feed, and WhatsApp-shareable image

5.2 Sample Grade 4 Weekly Timetable Structure

Day	Slot 1 (8:00)	Slot 2 (8:50)	Break	Slot 3 (10:10)	Slot 4 (11:00)	Slot 5 (11:50)
Monday	English (L&S — Pronunciation Drill)	Mathematics (Number Work)	Break	Kiswahili (Mazungumzo)	Social Studies (Theme intro)	Agriculture & Nutrition
Tuesday	Mathematics (New Concept)	English (Reading — Extensive)	Break	Integrated Science (Concept)	Creative Arts (Theory)	Religious Education
Wednesday	English (Language Use — Grammar)	Kiswahili (Sarufi)	Break	Integrated Science (PRACTICAL SESSION)	Integrated Science (PRACTICAL cont.)	Physical Education
Thursday	Mathematics (Problem Solving)	English (Writing — Guided Comp)	Break	Social Studies (Map/Activity work)	Agriculture (PRACTICAL)	Agriculture (PRACTICAL cont.)
Friday	Weekly Quiz / Assessment (All subjects)	Kiswahili (Utungo — Writing)	Break	Creative Arts (PRACTICAL project)	Reflection & Portfolio Update	Parent Progress Report

5.3 Practical Sessions — Subject-Specific Requirements

Subject	Grades	Duration	Activity Examples	Notes
Integrated Science	G4–G9	60–80 min	Simple home experiments (water filtration, plant growth, circuits); AI provides step-by-step narration; learner photographs result for portfolio	Safety checklist; parent supervision flag for chemical activities
Agriculture & Nutrition	G4–G6	60–90 min	Kitchen garden planting, soil testing, recipe preparation, food group identification	Seasonal planting calendar; Kenyan crop calendar integrated

Subject	Grades	Duration	Activity Examples	Notes
Pre-Technical Studies	G7–G9	90–120 min	Design-and-make projects using local materials; basic electrical/mechanical activities	Materials list generated; cost estimate in KES
Creative Arts & Sports	PP1–G12	40–60 min	Drawing, painting, music performance (sing/play), drama role-play, craft projects	Portfolio photo upload; peer/parent reviewer option
Physical & Health Ed	PP1–G12	40 min	Structured movement activities, sports skill drills, health demonstrations; AI provides narrated instructions	Works in small spaces; indoor alternatives for rain
Business Studies	G7–G9	60 min	Market survey activities, simple budgeting exercises, small enterprise simulation	Local market context (Kenyan prices, goods)
English Pronunciation	G1–G12	Within lesson	Tongue twisters, minimal pair drills, read-aloud passages, recorded reading assessments via STT	Whisper STT scoring; KICD target phoneme list per grade

6. Quizzes, Tests & Continuous Assessment

6.1 Assessment Types (CBC-Aligned) [+CBE Context]

The assessment system implements the three-tier CBC assessment model mandated by KNEC. All tracking is curriculum-centric: the system logs which learning outcomes each learner has mastered, displaying progress against CBC competencies rather than percentage scores.

Formative Assessment (Daily):

- End-of-lesson 2–3 question checks (oral or typed)
- Pronunciation scoring via STT (L&S strand)
- In-lesson activity completion tracking
- Socratic dialogue comprehension probes
- Not graded; used to update mastery profile
- Immediate AI feedback with CBC rubric language

Summative Assessment (SBA + National):

- Weekly Quiz: 10–15 min, covers week's SLOs
- End-of-theme Test: 20–30 min, all strands in theme
- End-of-term Assessment: 45–60 min, full-term SLOs
- Graded using 8-level KNEC scale (EE1–BE2)
- SBA records maintained for KPSEA/KJSEA (20% SBA contribution)
- Portfolio evidence compiled per term

6.2 Quiz Generation Specification

Question Type	Strand	Generation Rules	Code
Multiple Choice (MCQ)	All strands	4 options; 1 correct; 3 plausible distractors from common misconceptions in KICD guidance	MCQ
Fill-in-the-Blank	Language Use, Maths	Sentence with gap; accepts voice or typed input; determiners, tenses, prepositions etc.	FIB
Oral Response (Voice)	L&S strand, Kiswahili	Tutor asks question aloud; learner responds by voice; STT transcribes; AI evaluates against expected answer	ORL
Short Written Answer	Reading, Writing	2–5 sentence response; AI evaluates content, organisation, language use against CBC rubric	SWA
Practical Demonstration	Science, Agri, Arts	Learner photographs/videos practical output; parent/AI rates against practical rubric checklist	PRX
Composition Task	Writing strand	Guided composition prompt aligned to theme; AI evaluates narrative/functional/descriptive writing	COMP
Comprehension Passage	Reading strand	KICD-aligned reading passage with inferential and literal questions at increasing Bloom's levels	COMP_P

6.3 School-Based Assessment (SBA) Record

The platform automatically maintains KNEC-compliant SBA records for each learner. These contribute 20% to national assessments (KPSEA at Grade 6, KJSEA at Grade 9).

- SBA records stored per subject, per competency, per term

- Three SBA data points per term per subject (aligned to KNEC Grade 3, 4, 5, 7, 8 written test schedules)
- Parent can view, verify, and digitally acknowledge SBA records
- SBA export in KNEC-compatible format for submission to schools/exam centres
- Automatic alert when a learner's SBA trajectory puts KPSEA/KJSEA performance at risk

7. Progress Tracking, Portfolio & Longitudinal Analytics

7.1 Learner Mastery Profile (LMP) [+CBE Context]

Each learner has a continuously updated Learner Mastery Profile tracking competency attainment at the strand, sub-strand, and individual SLO level across all grades from PP1 through Grade 12. Dashboards highlight strong and weak areas, and early interventions can be flagged — all using the CBC/CBE descriptive scale.

- Competency map: 8-level score for each SLO in each subject, updated after every assessment
- Strand-level mastery: aggregate score per strand (L&S, Reading, Language Use, Writing for English)
- Subject-level mastery: overall subject rating per KNEC 8-point scale — reported without cross-subject aggregate
- CBC Core Competency tracker: which of the 7 competencies are developing
- Longitudinal growth chart: mastery scores plotted term-over-term from enrollment grade to current grade
- Prerequisite gap analysis: identifies which prerequisite competencies are missing for current-grade content
- Weakness heatmap: visual colour-coded grid of all strands and SLOs with performance levels
- Strength portfolio: curated collection of best-performance work samples, auto-tagged by competency

7.2 Progress Tracking Dashboards

Dashboard	Audience	Key Information	Update Frequency
Learner Dashboard	Student	Today's lesson progress bar; weekly streak; current mastery levels per subject; next recommended activity; milestone celebrations	Real-time
Parent Overview	Parent/Guardian	All children's progress at a glance; week summary; subjects needing attention; upcoming assessments; AI teaching tip for parent	Daily summary + real-time
Subject Deep-Dive	Parent/Guardian	Strand-by-strand mastery for each subject; SLO-level breakdown; growth trend chart; AI-recommended intervention if AE or BE	On demand
Timetable View	Parent/Guardian	Weekly timetable with completion status; missed lessons; rescheduling tool; practical session reminders	Weekly
Assessment History	Parent/Guardian	All quiz/test results with CBC rubric scores; SBA log; downloadable PDF reports per term	On demand
Longitudinal Report	Parent/Guardian	Grade-over-grade progress chart; CBC competency development across years; KPSEA/KJSEA readiness indicator	Termly
Insights & Alerts	Parent/Guardian	Weakness alerts (2+ consecutive AE/BE in a SLO); excellence celebrations (EE streak); AI intervention recommendations	Automated

7.3 Weakness Detection & Intervention Engine

- Pattern detection: if learner scores AE or BE on the same SLO in 2+ consecutive attempts, intervention is triggered
- Prerequisite trace: Knowledge Graph traversed to find the earliest unmastered prerequisite in the chain

- Remediation lesson: RAG retrieves prerequisite content and generates a targeted remediation mini-lesson
- Alternative pedagogy: second attempt uses a different explanation style (visual, example-based, real-world analogy)
- Parent notification: alert sent via WhatsApp/email with specific SLO, current level, and recommended action
- Intervention tracker: records which interventions were applied and their effectiveness; feeds back into personalisation

CBE Philosophy: Per KNEC/KICD CBE principles, no learner 'repeats' a grade for poor performance. The platform instead provides targeted support and alternative pathways, helping learners progress while building missing competencies. The intervention engine embodies this principle by offering multiple pedagogical approaches before escalating to parent/educator support.

8. Enhanced Multi-Agent Architecture

8.1 Agent Roster

Agent	Role	Core Function
Subject Matter Expert Agents (11)	Per subject per grade band	Generate CBC-aligned lesson content, activities, and assessments using RAG retrieval
Curriculum Alignment Agent	Cross-agent supervisor	Validates every output against retrieved KICD chunks; rejects and regenerates if alignment < 95%
Assessment Agent	Assessment lifecycle	Generates quizzes/tests; grades submissions against CBC rubric; maintains SBA records
Knowledge Tracing Agent	Mastery tracking	Updates Learner Mastery Profile after every interaction; Bayesian Knowledge Tracing model
Personalization Agent	Adaptive pacing	Adjusts difficulty, pacing, and content type based on LMP; selects next learning activity
Timetable Orchestrator Agent	Schedule generation	Creates and maintains weekly timetables; schedules practicals; handles reschedules
Voice Interaction Agent	Voice I/O	Manages STT/TTS pipeline; conducts pronunciation assessments; scores oral responses
Intervention Agent	Remediation	Detects weakness patterns; generates remediation pathways; notifies parents
Cross-Subject Integration Agent	Holistic learning	Identifies cross-subject connections from Knowledge Graph; surfaces integration opportunities
Parent Support Agent	Parent dashboard	Generates progress reports; provides parent teaching tips; sends WhatsApp/email alerts
Practical Session Agent	Hands-on learning	Generates practical activity guides; safety checklists; portfolio submission prompts
SBA Record Agent	KNEC compliance	Maintains SBA logs; generates KNEC-format reports; flags assessment deadlines

9. Recommended Technology Stack

This section is new in v2.1 Merged. It provides a production-grade, scalable, and secure technology stack recommendation optimised for the specific requirements of this platform: offline-first Kenyan deployment, child data protection, real-time voice AI, and a multi-agent LLM backend.

9.1 Frontend — Web & Mobile

Layer	Technology	Rationale
Framework	Next.js 14+ (App Router)	Server-side rendering for fast initial load on low-bandwidth connections; built-in API routes; excellent TypeScript support; Vercel deployment
Language	TypeScript (strict mode)	Type safety critical for complex data models (LMP, SBA records, CBC grading); reduces runtime errors in production
UI Component Library	shadcn/ui + Radix UI primitives	Accessible, headless components; full keyboard/screen-reader support; easily customisable for CBC branding
Styling	Tailwind CSS	Utility-first; consistent design tokens; small bundle size; responsive for mobile-first Kenyan users
State Management	Zustand (global) + TanStack Query (server state)	Lightweight global state for learner session; TanStack Query for caching, background sync, and offline-first data
Mobile App	React Native (Expo) with shared component library	Single codebase for iOS/Android; Expo Go for rapid testing; shared TypeScript types with web; supports offline mode
Voice UI	Custom React hooks wrapping Web Speech API / Whisper WASM	Push-to-talk UI component; waveform visualisation; progressive enhancement if STT unavailable
PWA Support	next-pwa + Workbox	Service worker caching of lessons for offline use; background sync when reconnected; installable on Android home screen
Animations	Framer Motion	Smooth, engaging transitions critical for young learners; milestone celebrations; progress indicators
Forms	React Hook Form + Zod	Type-safe form validation; critical for parent onboarding, SBA acknowledgment, consent flows
Charts & Visualisations	Recharts + D3.js (heatmaps)	Mastery heatmap, longitudinal growth charts, competency radar; accessible colour schemes
Internationalisation	next-intl	English/Kiswahili UI switching; right-to-left support ready; locale-aware date/number formatting

9.2 Backend — API & Business Logic

Layer	Technology	Rationale
API Framework	Next.js API Routes (primary) + FastAPI (Python microservices)	Next.js for BFF pattern; FastAPI for computationally heavy services (RAG, voice processing, timetable engine) where Python ML libraries are needed

Layer	Technology	Rationale
Runtime	Node.js 20 LTS (Next.js) + Python 3.11 (FastAPI)	LTS stability; Node for real-time and API gateway; Python for AI/ML processing
Authentication	NextAuth.js v5 (Auth.js) + JWT	Parent/learner/educator roles; magic link email login; Google OAuth; JWT for API auth; session stored in Supabase
API Design	tRPC (type-safe API) for internal + REST for external	tRPC eliminates API contract drift between frontend and backend; REST for third-party integrations (WhatsApp, KNEC)
Real-time	Supabase Realtime (WebSockets)	Live lesson progress updates; voice session state sync; parent dashboard real-time alerts without polling
Background Jobs	BullMQ + Redis	Async content generation jobs (lesson factory, timetable generation); assessment grading queue; parent report generation
Caching	Redis (Upstash serverless)	Cache generated lessons (expensive RAG operations); learner session state; quiz bank; leaderboard/streak data
File Processing	Sharp (image), FFmpeg (audio/video)	Compress practical session photos before portfolio upload; process voice recordings for STT

9.3 AI & ML Stack

Component	Technology	Rationale
Primary LLM	Claude claude-sonnet-4-20250514 (Anthropic API)	Best-in-class instruction following; strong multilingual (English/Kiswahili); long context for curriculum chunks; safe for children
LLM Orchestration	LangChain (Python) + LangGraph for multi-agent	Agent graph orchestration; tool use; memory management; streaming responses; well-supported with Weaviate
Vector Database	Weaviate Cloud (production) / Weaviate local (dev)	Purpose-built for hybrid search (BM25 + vector); built-in multi-tenancy (per-learner isolation); strong Python/JS clients
Embeddings	multilingual-e5-large (via sentence-transformers)	Best multilingual embedding for English+Kiswahili; runs locally; free; strong semantic retrieval for curriculum text
Knowledge Graph	Neo4j Aura Free (dev) / Neo4j Enterprise (prod)	Native graph queries for prerequisite chains; cross-subject links; Cypher query language; REST and Bolt drivers
STT (primary)	Whisper (whisper.cpp for local / OpenAI API for cloud)	Highest accuracy for Kenyan-accented English and Kiswahili; offline via whisper.cpp; WASM build for browser
STT (fallback/offline)	Vosk (50MB model, fully offline)	Runs on low-end Android devices; free; no data sent to cloud — critical for child privacy
TTS (primary)	Kokoro-82M (local, open-source)	High quality; runs on CPU; free; no API cost at scale; supports custom voice training
TTS (fallback)	Coqui TTS (open-source)	Kiswahili voice model available; fully offline; free
Knowledge Tracing	PyBKT (Bayesian Knowledge Tracing)	Well-validated academic model for learner mastery probability per skill; open-source Python library

Component	Technology	Rationale
Recommendation	Custom collaborative filtering + content-based (Surprise library)	Next activity recommendation based on mastery gaps and peer performance patterns

9.4 Database & Storage

Component	Technology	Rationale
Primary Database	Supabase (PostgreSQL)	Row-level security for multi-tenant learner data; built-in auth; real-time subscriptions; KNEC-compatible structured SBA records; free tier generous
Database ORM	Prisma (TypeScript-first)	Type-safe database queries; auto-generated types shared with frontend; migration management; works perfectly with Supabase/PostgreSQL
Vector Store	Weaviate	See AI stack above
Graph Database	Neo4j	See AI stack above
Portfolio Storage	Cloudflare R2	S3-compatible; $0.015/GB after 10GB free; zero egress fees critical for Kenyan bandwidth costs; automatic CDN via Cloudflare
Session/Cache Store	Redis (Upstash)	Serverless Redis; per-request billing; low latency; BullMQ job queue support
Search (full-text)	Meilisearch (self-hosted) or Typesense	Fast full-text search over lesson content, SBA records, parent notes; Kenyan-hosted for latency

9.5 Infrastructure & DevOps

Component	Technology	Rationale
Frontend Hosting	Vercel (Next.js)	Zero-config Next.js deployment; edge network; automatic preview deployments; free tier for MVP
Backend Services	Railway or Render (FastAPI microservices)	Simple Python service deployment; auto-scaling; lower cost than AWS for early stage; easy migration to AWS later
Container Orchestration	Docker Compose (dev) → Kubernetes/EKS (production scale)	Containerise all services; Docker Compose for local dev; migrate to K8s when user volume justifies it
CI/CD	GitHub Actions	Automated testing, linting, type checking on every PR; deploy previews; secret scanning
Monitoring	Sentry (errors) + PostHog (product analytics) + Grafana (infra)	Sentry for error tracking; PostHog for learner behaviour analytics (privacy-first, self-hostable); Grafana for RAG/voice latency monitoring
Secret Management	Doppler or Infisical (open-source)	Centralised secrets management; environment-specific configs; audit trail; critical for API key rotation
CDN	Cloudflare	Kenya edge PoP; DDoS protection; WAF; R2 storage integration; free tier substantial
Email	Resend (transactional)	Parent notifications; SBA reports; clean API; good deliverability; built on React Email templates

Component	Technology	Rationale
Push Notifications	Firebase Cloud Messaging (free tier)	Android-first (dominant in Kenya); parent lesson reminders; weakness alerts; assessment deadline notifications
WhatsApp Integration	WhatsApp Business API (Meta) via 360dialog	Progress reports to parent phones; works without smartphone app; reach parents on feature phones
Logging	Pino (Node.js) + structlog (Python) → Loki + Grafana	Structured JSON logging; low overhead; centralised log aggregation; RAG citation audit trail

9.6 Security Architecture

Area	Approach	Implementation
Authentication	Multi-factor for parent accounts; magic link for learners	NextAuth.js v5; TOTP 2FA for parent admin; email magic links for child-safe login
Authorisation	Role-based access control (RBAC)	Supabase Row Level Security; parent can only access own children's data; educator scoped to assigned learners
Data Encryption	Encryption at rest + in transit	Supabase encrypted at rest (AES-256); TLS 1.3 everywhere; Cloudflare R2 server-side encryption for portfolio files
Child Data Protection	Kenya Data Protection Act 2019 + COPPA principles	No voice recordings stored without written parental consent; data minimisation; 30-day deletion on account closure; no third-party data sharing
API Security	Rate limiting + input validation + CORS	Upstash rate limiting middleware; Zod input validation on all endpoints; strict CORS; helmet.js headers
Dependency Security	Automated vulnerability scanning	Dependabot + Snyk in CI/CD pipeline; weekly dependency audits; no known CVEs in production
Audit Logging	Immutable audit trail for SBA records	All SBA record writes logged with timestamp, method, and agent ID; parent acknowledgment stored with digital signature hash
Penetration Testing	Annual third-party pen test	Required before commercial launch; OWASP Top 10 coverage; child data protection specific test cases

9.7 Voice Pipeline — Latency Targets

Input Event	Output Event	Target Latency
Push-to-talk activated	Voice capture starts	< 200ms (UI response)
Speech recording ends	STT transcription complete	< 1.5s (Whisper local, 10-sec utterance)
STT output	RAG retrieval + LLM response	< 2.5s (cached lesson context)
LLM response text	TTS audio starts playing	< 800ms (Kokoro local TTS)
Total voice-in to voice-out	Full conversational round-trip	< 5 seconds (target); < 8s (acceptable)

10. Updated Implementation Roadmap

Phase 1: Foundation + Grade 4 English MVP (Weeks 1–8)

- KICD Grade 4 English CD fully ingested: all 13 themes, 4 strands, ~150 lessons chunked and embedded
- RAG pipeline operational: hybrid retrieval (BM25 + dense) + Curriculum Alignment Agent at 95%+ accuracy
- Daily lesson generator: produces 40-min structured lessons for all 13 Grade 4 English themes
- Weekly timetable engine: auto-generates English timetable; exports PDF + iCal
- Voice AI MVP: TTS reads all lesson content; STT for pronunciation drills (L&S strand)
- Assessment engine: end-of-lesson checks, weekly quizzes, end-of-theme tests using KICD rubrics
- CBC 8-level grading: all assessments output EE1–BE2 descriptors
- Parent dashboard: lesson completion, weekly quiz scores, weakness alerts
- Offline support: lessons cached for offline use; sync on reconnect
- Beta test: 20 Grade 4 families using English subject

Phase 2: Grade 4 All Subjects (Weeks 9–16)

- Ingest all Grade 4 subject CDs: Kiswahili, Mathematics, Integrated Science, Social Studies, Creative Arts, Agriculture & Nutrition, CRE/IRE/HRE
- 11 Subject Matter Expert Agents deployed and tested
- Full multi-subject timetable generator with practical session scheduling
- Practical Session Agent: home experiment guides for Science and Agriculture
- Conversational voice tutor: full bidirectional dialogue across all subjects
- SBA record system: KNEC-format records maintained; parent acknowledgment workflow
- Cross-subject Knowledge Graph: integrate prerequisites and cross-subject links for Grade 4
- 50 families pilot across all Grade 4 subjects

Phase 3: Grades 1–6 Expansion (Weeks 17–28)

- Ingest all KICD CDs for Grades 1–3 (Lower Primary) and Grades 5–6 (Upper Primary)
- Longitudinal Learner Mastery Profile: PP1 through Grade 6 tracking
- KPSEA readiness dashboard: Grade 5–6 SBA trajectory and Grade 6 preparation mode
- Advanced Knowledge Tracing: Bayesian KT model for mastery probability per SLO
- Predictive intervention: proactive weakness detection 2–3 weeks before assessment
- 200+ families enrolled; full primary (Grade 1–6) operational

Phase 4: Junior & Senior Secondary (Weeks 29–48)

- Ingest all KICD CDs for Grades 7–9 (Junior Secondary) — all 9 subjects
- KJSEA preparation module: Grade 8–9 intensive assessment preparation; 8-level grading simulation
- Senior Secondary (Grades 10–12): track selection engine (STEM / Social Sciences / Arts & Sports / VT)
- Track-specific Subject Agents deployed for each Senior School pathway
- 500+ families; full PP1–G12 capability operational

11. Success Metrics & KPIs

11.1 RAG & Content Quality Metrics

Metric	Target	Measurement Method
Curriculum Adherence Rate	95%+ Month 1; 97%+ Month 6	Automated: semantic similarity of output to KICD source chunks
RAG Retrieval Precision@k	> 0.80 at k=5	Monthly: human review of 100 generated lessons vs. source chunks
Hallucination Rate	< 2%	Monthly audit: 50 lessons reviewed by CBC curriculum specialist
SLO Coverage per Theme	100% — all SLOs addressed within theme	Automated: SLO checklist per generated theme block
Voice STT Accuracy (Children)	> 85% word accuracy	Weekly: 100 voice interactions reviewed; Kenyan child speech test set
TTS Naturalness Score	> 4.0 / 5.0 (parent/student survey)	Monthly parent/student satisfaction survey on voice quality

11.2 Learning Outcome Metrics

Metric	Target	Measurement Method
SLO Mastery Rate (ME+ level)	70% at 6mo; 80% at 12mo	Rubric assessment per SLO across all enrolled learners
Student Learning Gains	15%+ at 6mo; 20%+ at 12mo	Pre/post assessment per term; competency-mapped
Competency Development (CBC 7)	5/7 by 6mo; 7/7 by 12mo	Portfolio analysis by CBC curriculum specialist
Practical Completion Rate	> 60% of scheduled practicals completed	Practical session agent logs; photo submission rate
Parent Teaching Time Saved	> 50% at 6mo	Parent survey; time-tracking data
KPSEA Readiness (Grade 6)	> 70% learners at ME or above	End-of-G5 benchmark assessment simulating KPSEA

12. Privacy, Security & CBC Compliance

12.1 Child Data Protection [+CBE Context]

- Kenya Data Protection Act (2019): full compliance for all personal data of minors
- COPPA and FERPA-aligned principles: parent access rights, no third-party sharing of educational records
- Voice data: STT processing local-first; no child voice recordings stored without explicit written parental consent
- Portfolio photos/videos: stored encrypted; accessible only to learner, parent, and assigned educator
- Data minimisation: only data necessary for learning and assessment is collected
- Right to deletion: all learner data deleted within 30 days of account closure request
- GDPR alignment: consent management, data portability, breach notification within 72 hours

12.2 CBC/CBE Curriculum Compliance

- All KICD curriculum documents licensed for use: written authorisation from KICD required before commercial launch
- Curriculum version control: platform tracks KICD curriculum design version; re-ingestion triggered on any KICD update
- Annual CBC educator review: 2 certified CBC curriculum specialists review 200+ lessons per year
- Rationalisation alignment: platform reflects 2024 KICD subject rationalisation (max subjects per level)
- No summative ranking: platform explicitly prohibits display of aggregate scores or learner rankings, per KNEC CBAF
- Assessment integrity: SBA records maintain audit trail with timestamp and generation method metadata

13. Gaps Addressed from Previous Versions

Gap / Version	Resolution
v1.0: RAG described at high level only	Full 8-step RAG pipeline specified (Section 2); chunk hierarchy, metadata schema, retrieval strategy, validation gates all defined
v1.0: No curriculum content generation pipeline	Lesson Factory specified (Section 3): daily→weekly→termly→annual content pipeline; Grade 4 English as reference model
v1.0: Voice functionality not specified	Full bidirectional voice architecture (Section 4): STT (Whisper + Vosk), TTS (Kokoro + Coqui), 7 dialogue mode patterns, child safety, latency targets
v1.0: No timetable/scheduling system	Timetable Engine specified (Section 5): multi-subject weekly timetable, KICD lesson allocation rules, practical session scheduling, iCal/WhatsApp export
v1.0: Practical sessions not specified	Practical Session Agent defined (Section 5.3): all applicable subjects, duration, activity examples, materials lists in KES, safety checklists
v1.0: CBC grading not fully detailed	KNEC 8-level CBAF grading fully implemented (Section 1.3): EE1–BE2 with score ranges, points, AI actions; no aggregate scores per KNEC directive
v1.0: Progress tracking lacked competency granularity	Learner Mastery Profile (Section 7): SLO-level tracking, competency development per CBC 7 core competencies, longitudinal PP1–G12 tracking
v1.0: No SBA/KPSEA/KJSEA integration	SBA Record Agent (Section 8); KPSEA readiness dashboard (Phase 3); KJSEA preparation module (Phase 4); SBA export in KNEC format
v1.0: Senior Secondary underdeveloped	Track selection engine for all 4 pathways (STEM/SS/Arts/VT); track-specific agents; track-aware timetable generation
v1.0: No weakness detection	Intervention Engine (Section 7.3): pattern detection, prerequisite trace, alternative pedagogy, parent notification, intervention tracking
v2.0: No tech stack recommendation	Full production-grade stack specified (Section 9): Next.js/TypeScript frontend, FastAPI/Python AI backend, Weaviate/Neo4j/Supabase data layer, security architecture
v2.0: Kenyan deployment constraints underspecified	Offline-first architecture (Vosk/Coqui), PWA/Expo mobile, Cloudflare CDN for Kenya edge, WhatsApp-first parent communication, low-bandwidth voice caching
v2.0: Child privacy compliance detail lacking	Full compliance framework (Section 12): KDPA 2019, COPPA, FERPA, GDPR alignment; local-first voice processing; consent management

14. Immediate Next Steps (Week 1)

1. Obtain KICD curriculum documents for Grade 4 all subjects — source official PDFs from kicd.ac.ke or authorised distributors. The Grade 4 English CD is the first reference model for the RAG chunking pipeline.

2. Establish KICD authorisation — contact KICD for a formal licence to use curriculum content in an AI platform. Required before commercial launch.

3. Set up RAG infrastructure — deploy Weaviate (local) + Neo4j Community + sentence-transformers. Ingest Grade 4 English CD using the 5-level chunking hierarchy defined in Section 2.2.

4. Scaffold Next.js + TypeScript project — initialise monorepo (Turborepo), configure ESLint/Prettier/Zod, set up Supabase instance, configure NextAuth.js with parent/learner roles.

5. Build Lesson Factory MVP — implement the daily lesson template (Section 3.2) and test generation across all 13 Grade 4 English themes. Validate against KICD SLOs.

6. Integrate Whisper STT + Kokoro TTS — set up voice pipeline; test with Kenyan children's English and Kiswahili pronunciation drills (Grade 4 L&S strand phoneme targets).

7. Implement CBC 8-level grading engine — build the assessment scoring module using the KNEC EE1–BE2 scale; ensure no aggregate score computation.

8. Build timetable generator prototype — generate a Grade 4 weekly timetable using Section 5.2 as template; validate lesson counts against KICD allocation.

9. Recruit CBC curriculum specialist advisor — minimum 1 KICD-trained educator for monthly content audits starting from Week 6.

10. Configure CI/CD pipeline — GitHub Actions with TypeScript type-checking, Zod schema validation, Prisma migration safety checks, and Sentry error tracking from Day 1.

This platform has the potential to democratise high-quality, personalised CBC/CBE education for every homeschooling family in Kenya — from Pre-Primary through Senior Secondary — while maintaining strict curriculum fidelity and preparing learners for KPSEA and KJSEA national assessments.Document: CBC HomeLearning Platform PRD v2.1 Merged | March 2026 | Confidential Draft